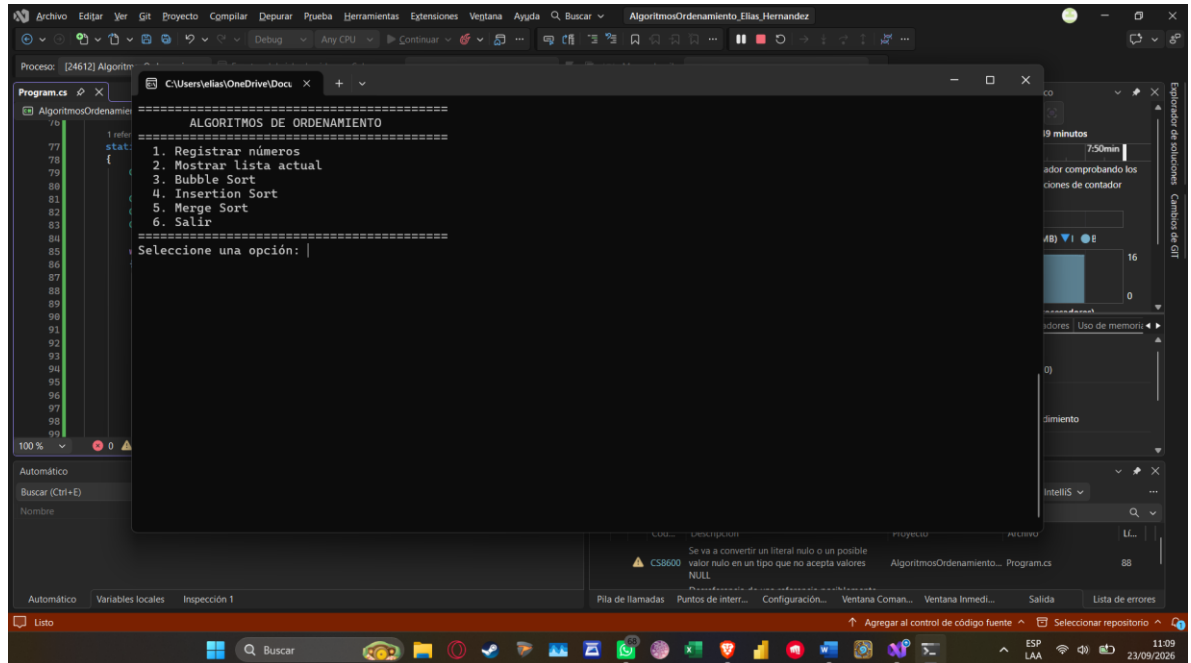
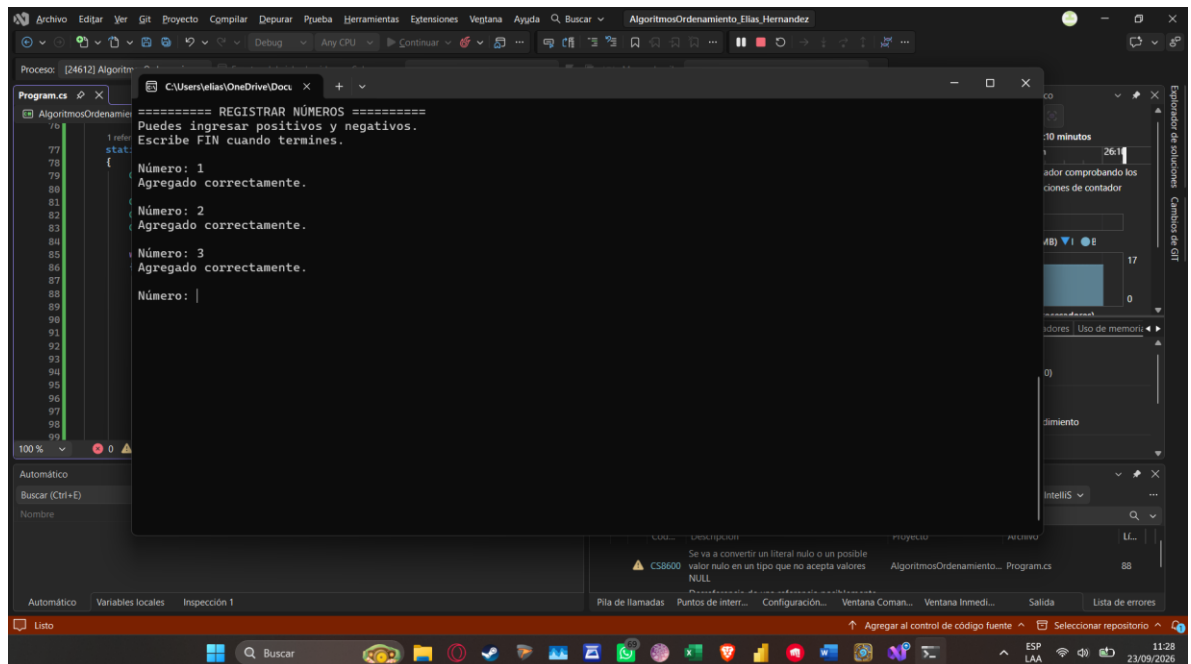


Manual de uso Algoritmos de ordenamiento

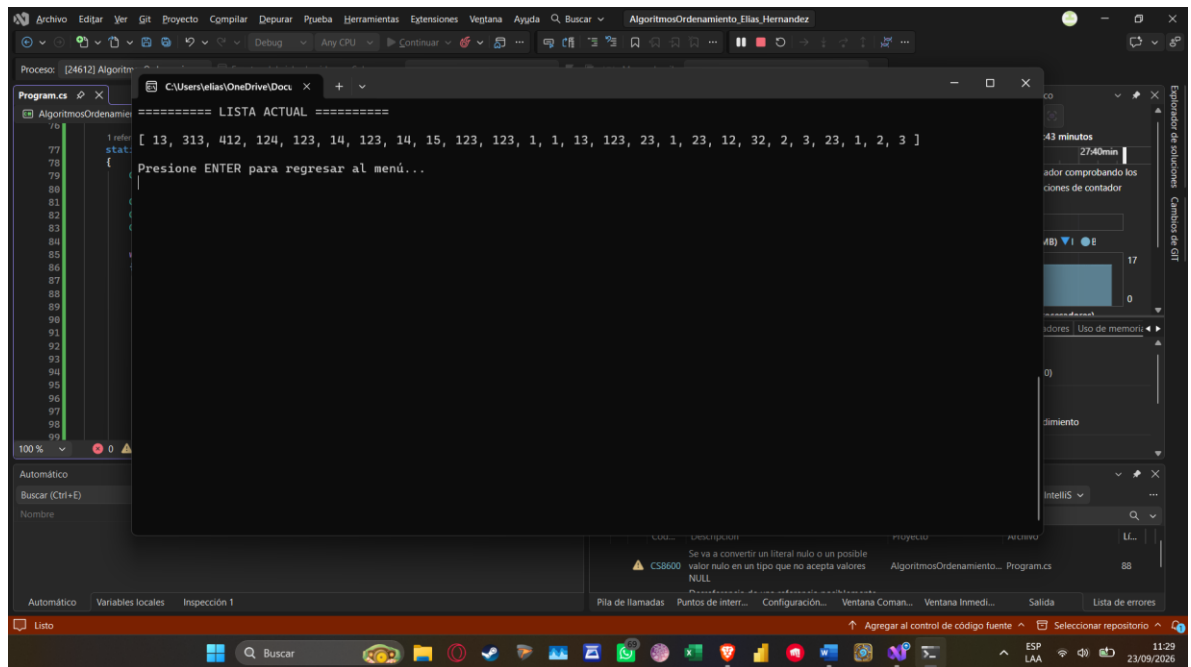
1. El menú consta de 6 opciones.



2. La primera opción es Registrar números donde podremos ingresar números hasta que escribamos fin.



3. La segunda opción es Mostrar lista actual.



The screenshot shows the Visual Studio IDE with a console window displaying the output of a program. The output is titled "LISTA ACTUAL" and shows a list of numbers: [13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3]. Below the list, it says "Presione ENTER para regresar al menú...". The IDE interface includes a menu bar, a toolbar, a solution explorer, and a status bar.

```
Proceso: [24612] Algoritmo
C:\Users\elias\OneDrive\Docu...
Program.cs
AlgoritmosOrdenamiento
76
77
78
79
80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100 %
Automático
Buscar (Ctrl+E)
Nombre
```

LISTA ACTUAL

```
[ 13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3 ]
Presione ENTER para regresar al menú...
```

Se va a convertir un literal nulo o un posible valor nulo en un tipo que no acepta valores NULL

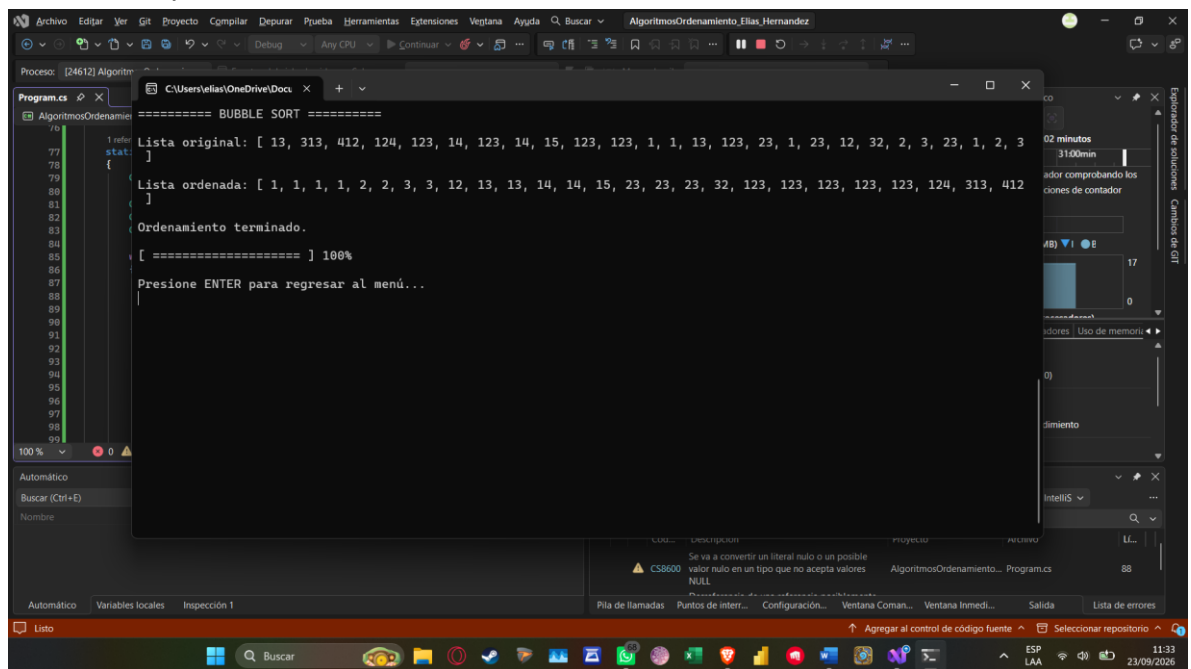
AlgoritmosOrdenamiento... Program.cs 88

Variables locales Inspección 1

Agregar al control de código fuente Seleccionar repositorio

11:29 23/09/2026

4. La tercera opción es Bubble Sort.



The screenshot shows the Visual Studio IDE with a console window displaying the output of a program. The output is titled "BUBBLE SORT" and shows the original list: [13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3]. Below the list, it says "Lista ordenada: [1, 1, 1, 1, 2, 2, 3, 3, 12, 13, 13, 14, 14, 15, 23, 23, 23, 32, 123, 123, 123, 123, 124, 313, 412]". Below the ordered list, it says "Ordenamiento terminado." and "[=====] 100%". Below the percentage, it says "Presione ENTER para regresar al menú...". The IDE interface includes a menu bar, a toolbar, a solution explorer, and a status bar.

```
Proceso: [24612] Algoritmo
C:\Users\elias\OneDrive\Docu...
Program.cs
AlgoritmosOrdenamiento
76
77
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81
82
83
84
85
86
87
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89
90
91
92
93
94
95
96
97
98
99
100 %
Automático
Buscar (Ctrl+E)
Nombre
```

BUBBLE SORT

```
Lista original: [ 13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3 ]
Lista ordenada: [ 1, 1, 1, 1, 2, 2, 3, 3, 12, 13, 13, 14, 14, 15, 23, 23, 23, 32, 123, 123, 123, 123, 124, 313, 412 ]
Ordenamiento terminado.
[ ===== ] 100%
Presione ENTER para regresar al menú...
```

Se va a convertir un literal nulo o un posible valor nulo en un tipo que no acepta valores NULL

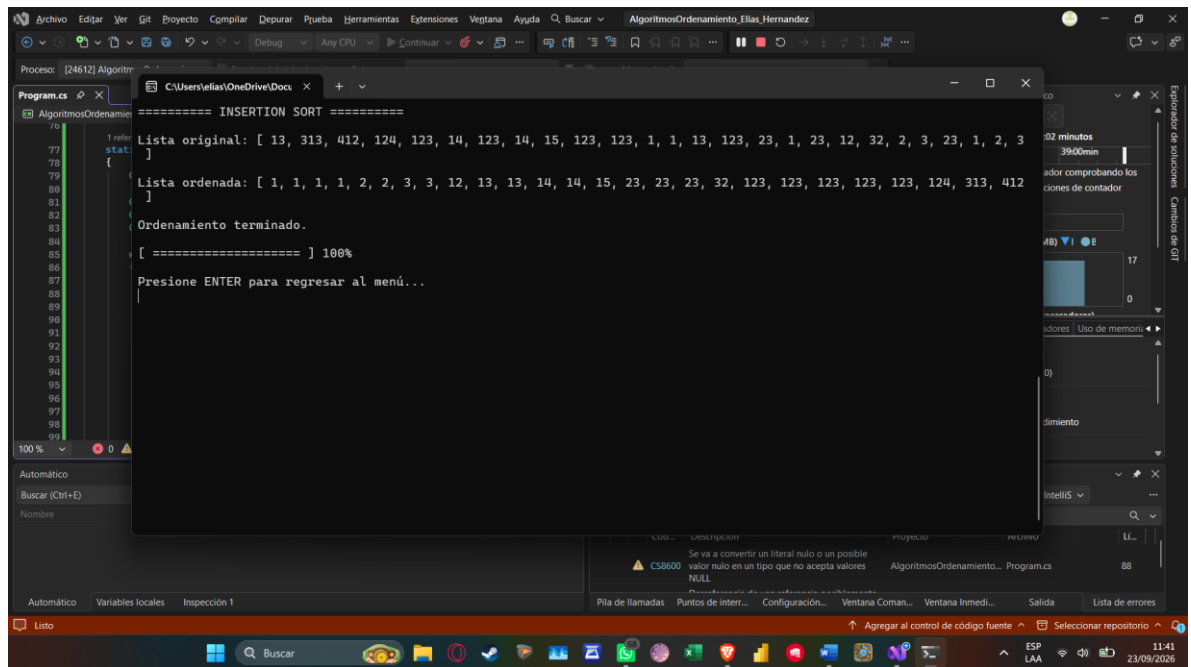
AlgoritmosOrdenamiento... Program.cs 88

Variables locales Inspección 1

Agregar al control de código fuente Seleccionar repositorio

11:30 23/09/2026

5. La cuarta opción es Insertion Sort.

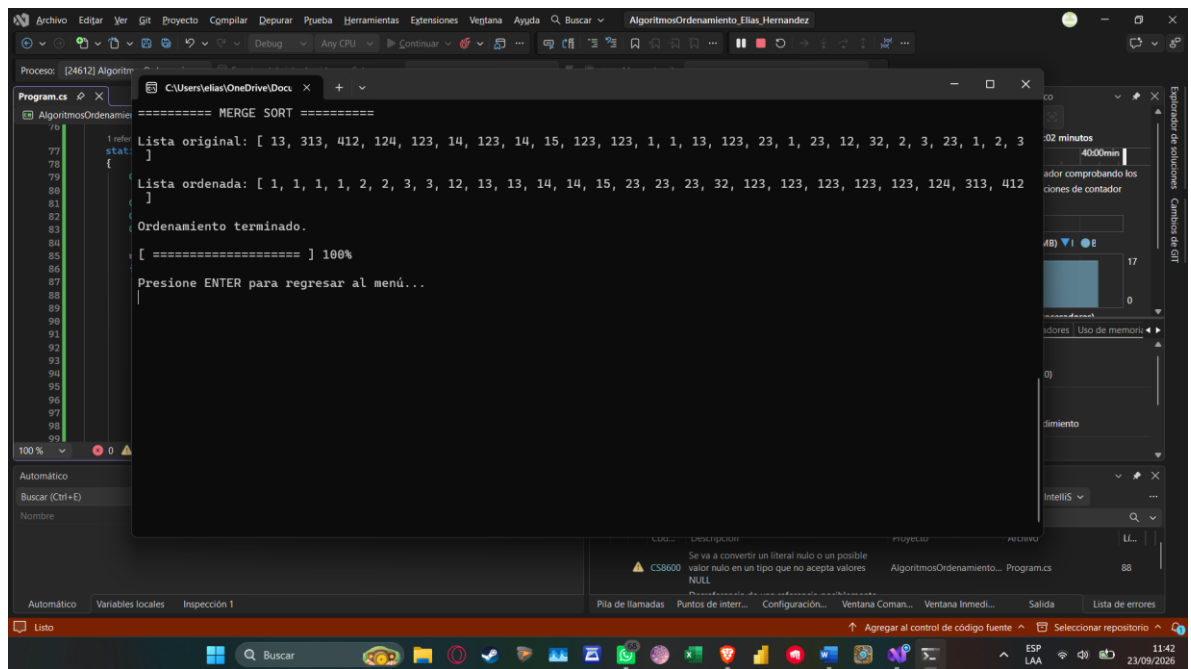


The screenshot shows the Visual Studio IDE with a C# program implementing Insertion Sort. The code is as follows:

```
Program.cs
70
71
72 ===== INSERTION SORT =====
73
74 Lista original: [ 13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3 ]
75
76 Lista ordenada: [ 1, 1, 1, 1, 2, 2, 3, 3, 12, 13, 13, 14, 14, 15, 23, 23, 23, 32, 123, 123, 123, 123, 123, 124, 313, 412 ]
77
78 Ordenamiento terminado.
79
80 [ ===== ] 100%
81
82 Presione ENTER para regresar al menú...
83
84
85
86
87
88
89
90
91
92
93
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95
96
97
98
99
```

The output window shows the execution of the program, displaying the original and sorted lists, and the completion status.

6. La quinta opción es Merge Sort.

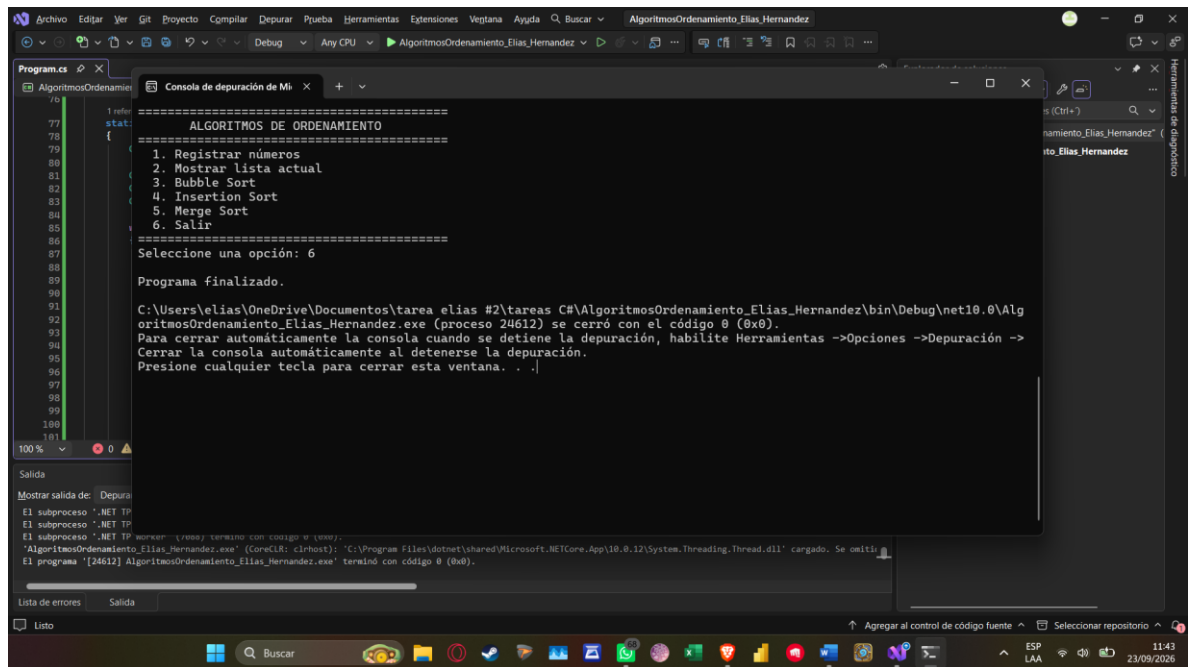


The screenshot shows the Visual Studio IDE with a C# program implementing Merge Sort. The code is as follows:

```
Program.cs
70
71
72 ===== MERGE SORT =====
73
74 Lista original: [ 13, 313, 412, 124, 123, 14, 123, 14, 15, 123, 123, 1, 1, 13, 123, 23, 1, 23, 12, 32, 2, 3, 23, 1, 2, 3 ]
75
76 Lista ordenada: [ 1, 1, 1, 1, 2, 2, 3, 3, 12, 13, 13, 14, 14, 15, 23, 23, 23, 32, 123, 123, 123, 123, 123, 124, 313, 412 ]
77
78 Ordenamiento terminado.
79
80 [ ===== ] 100%
81
82 Presione ENTER para regresar al menú...
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
```

The output window shows the execution of the program, displaying the original and sorted lists, and the completion status.

7. La sexta opción es salir.



https://github.com/Okami7u7/AlgoritmosOrdenamiento_Elias_Hernandez